WORKED SOLUTION

STEP III 2001, Question 5

Pure • Normals to a Parabola

STEP III 2001, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q5

The key idea

Asking which normals pass through a fixed point (h, k) turns into asking how many real roots a cubic in the parameter t has. The cubic has no t^2 term, so it has exactly the shape $t^3 + pt + q$ from the opening part. The number of real roots changes only when two roots merge into a repeated root, so the boundary between "one normal" and "three normals" is where the cubic has a repeated root. That boundary is the locus we want.

Solution

The cubic

Let $f(x) = x^3 + px + q$. Then $f'(x) = 3x^2 + p \geq 0$ for all x when $p \geq 0$, and $f'(x) = 0$ can happen at most at the single point $x = 0$ (only when $p = 0$). So f is strictly increasing on $\mathbb{R}$: if $x_1 < x_2$ then $f(x_2) - f(x_1) = \int_{x_1}^{x_2} f'(x)\,\mathrm{d}x > 0$, because the integrand is non-negative and zero at no more than one point.

Also $f(x) \to -\infty$ as $x \to -\infty$ and $f(x) \to +\infty$ as $x \to +\infty$. Since f is continuous, it takes the value 0 somewhere, and because it is strictly increasing it does so only once. So there is exactly one real solution.

The normals through (h, k)

On C, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t} = \dfrac{2a}{2at} = \dfrac{1}{t}$, so the normal at the point with parameter t has gradient $-t$. (This also works at $t = 0$, where the tangent is vertical and the normal is the x-axis.) The normal is

$$y - 2at = -t\,(x - at^2), \qquad \text{that is} \qquad y + tx = 2at + at^3.$$

It passes through (h,k) exactly when $k+th = 2at + at^3$, i.e.

$$\boxed{at^3 + (2a-h)\,t - k = 0} \qquad \text{or, dividing by } a>0, \qquad t^3 + \frac{2a-h}{a}\,t - \frac{k}{a} = 0.$$

Ask yourself Could two different values of t give the same normal line, so that counting roots overcounts normals?

No. The normal at t has gradient $-t$, so different values of t give normals with different gradients, and these are different lines. So the number of normals through (h,k) is exactly the number of distinct real roots of the cubic.

If $h \le 2a$ then the coefficient $p = \dfrac{2a-h}{a}$ is ≥ 0, so by the first part the cubic has exactly one real root. Hence exactly one normal to C passes through (h,k), as required.

The locus of points with exactly two normals

Write the cubic as $t^3 + Pt + Q = 0$ with $P = \dfrac{2a-h}{a}$ and $Q = -\dfrac{k}{a}$. A real cubic has either one or three real roots counted with multiplicity. So it has exactly two *distinct* real roots exactly when it has a double root s and a different simple root.

The roots add up to 0, since there is no t^2 term, so the simple root is $-2s$. We need $s \neq 0$ so that $-2s \neq s$. Comparing coefficients,

$$t^3 + Pt + Q = (t-s)^2(t+2s) = t^3 - 3s^2 t + 2s^3,$$

so $P = -3s^2$ and $Q = 2s^3$. Translating back:

$$h = 2a + 3as^2, \qquad k = -2as^3, \qquad s \neq 0.$$

To eliminate s, note $(h-2a)^3 = 27a^3s^6$ and $k^2 = 4a^2s^6$, so $4(h-2a)^3 = 108a^3s^6 = 27a\,k^2$.

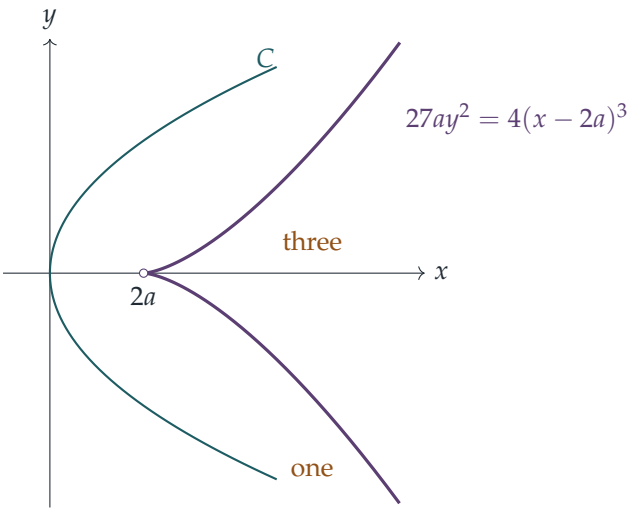
Conversely, suppose $27ak^2 = 4(h-2a)^3$ with $(h,k) \neq (2a,0)$. Then $h - 2a \ge 0$, and in fact $h > 2a$ (if $h = 2a$ then $k = 0$). Let s be the real number with $2as^3 = -k$. Then $4a^2s^6 = k^2 = \frac{4}{27a}(h-2a)^3$, so $(h-2a)^3 = (3as^2)^3$ and therefore $h - 2a = 3as^2$. So (h,k) has the form above with $s \neq 0$, and exactly two normals pass through it. (Equivalently, this is the repeated-root condition $4P^3 + 27Q^2 = 0$.)

The point $(2a,0)$ itself is excluded. There $P = Q = 0$, the cubic is $t^3 = 0$ with a triple root, and only one normal (the x-axis) passes through it.

So the required locus is

$$\boxed{27a\,y^2 = 4\,(x-2a)^3}, \qquad \text{excluding the point } (2a,0).$$

This is a semicubical parabola with a cusp at $(2a,0)$, symmetric about the x-axis and lying in $x \ge 2a$, with $y = \pm\dfrac{2}{3\sqrt{3a}}(x-2a)^{3/2}$. Inside it (the region containing the positive x-axis beyond $2a$) the cubic has three distinct real roots, so three normals pass through each point. Outside it only one normal does.



The sketch uses $a = 1$. The open circle marks the excluded cusp, and the labels give the number of normals through points in each region.

Checks

- Take $a = 1$ and $s = 1$, which gives the point $(5, -2)$. The cubic is $t^3 - 3t + 2 = (t-1)^2(t+2)$, so the normals at $(1, 2)$ and $(4, -4)$ both pass through $(5, -2)$. Indeed, the normal at $t = 1$ is $y + x = 3$ and the normal at $t = -2$ is $y - 2x = -12$. Both contain $(5, -2)$, and $27 \cdot 4 = 108 = 4 \cdot 27$.
- At $(5, 0.1)$ (with $a = 1$), $4P^3 + 27Q^2 = -108 + 0.27 < 0$, giving three normals. At $(5, 4)$ it is $-108 + 432 > 0$, giving one normal. This matches the sketch.
- We tested the normal equation and the one-root claims numerically on hundreds of random cases.

Common slips

- Forgetting to exclude the cusp $(2a, 0)$. The equation is satisfied there, but there is only one normal.
- Claiming $f'(x) > 0$ when $p \geq 0$. For $p = 0$, $f'(0) = 0$, so you need the slightly stronger argument that f is still strictly increasing.
- Writing the gradient of the normal as $1/t$ (the tangent's gradient) instead of $-t$.

Beyond STEP

The locus is the *evolute* of the parabola: the curve traced out by its centres of curvature, and also the envelope of its normals. Crossing the evolute changes the number of

normals from a point by two, which is a first example of the ideas behind catastrophe theory and the discriminant of a cubic. Siklos, *Advanced Problems in Mathematics*, has more STEP problems in this spirit. Bruce and Giblin, *Curves and Singularities*, develops evolutes and envelopes properly.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.